

Research on Student Classroom Behavior Recognition Based on YOLOv11

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Abstract

To address the issues of weak multi-scenario generalization capability and insufficient small-target detection accuracy in classroom behavior recognition, this study proposes an improved YOLOv11 model. First, a Multi-Scenario Classroom Behavior Dataset (MSCB-Dataset) was constructed, encompassing three typical classroom scenarios and seven core behavior categories, comprising 5,709 images. Second, the backbone network was replaced with Swin Transformer to capture both global and local features, and an EMASlide-Loss function was designed to optimize the training performance for small targets. Finally, model lightweighting was achieved through INT8 quantization and structured pruning.

Experimental results demonstrate that the improved model achieves an mAP of 87.6%, representing a 4.8% improvement over the original YOLOv11. After lightweighting, the model parameters are reduced by 42.3%, with inference latency ≤ 30 ms, meeting the deployment requirements for edge devices. This study provides practical reference for the engineering application of classroom behavior recognition technology.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision.

Keywords

Smart Education, Classroom Behavior Recognition, YOLOv11, Swin Transformer, Model Lightweighting

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1 INTRODUCTION

Mainstream approaches to classroom behavior recognition can be categorized into three types: sensor-based, audio analysis-based, and computer vision-based methods. Among these, computer vision-based approaches utilizing body movements have become the research focus due to their compatibility with fixed classroom camera acquisition scenarios.



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Tan et al. [1] designed an anomaly detection system that did not incorporate human pose features, resulting in poor recognition robustness under occluded conditions. Tang et al. [2] proposed an improved YOLOv5-based algorithm for student classroom behavior recognition, which effectively enhanced the recognition capability for occluded targets. Zhang et al. [3] implemented a classroom behavior recognition model based on the SlowFast network, achieving an mAP of 82.4%; however, the model parameters exceeded 50M with inference latency surpassing 100 ms, making it difficult to meet the real-time monitoring requirements of classroom settings. Yan et al. [4] integrated YOLOv5 with BetaPose to achieve recognition of five classroom behavior categories, but the miss detection rate exceeded 15% in occluded interaction scenarios. Liu et al. [5] proposed the YOLOv8n_BT model, which achieved an mAP of 83.7% in occluded scenes, yet cross-scenario generalization validation was not conducted. Zhang et al. [6] presented a classroom behavior recognition algorithm combining super-resolution and object detection, employing SRGAN to enhance image resolution and improving YOLOv8 to strengthen recognition of small and densely packed targets. Liu et al. [7] proposed an improved YOLOv8n_BT algorithm, effectively enhancing the behavior recognition capability for occluded targets and small targets in the back rows.

In summary, existing research exhibits evident limitations in cross-scenario generalization, recognition accuracy for occluded targets, and the balance between real-time performance and lightweight design. Therefore, this study proposes improvements to the backbone network and loss function of YOLOv11 [8], employs quantization and pruning techniques for model lightweighting, and conducts multi-dimensional validation of model performance.

2 EXPERIMENTAL DATASET

This study constructs a Multi-Scenario Classroom Behavior Dataset (MSCB-Dataset) for model training to validate the feasibility of classroom behavior recognition methods. The dataset was collected in situ via classroom cameras and includes seven core behavior categories: listening attentively, writing, conversing, dozing off, using a mobile phone, standing up, and raising a hand. The selection of these seven categories establishes a comprehensive recognition framework for core classroom behaviors. This framework is designed based on the following logic to ensure analytical completeness: “listening attentively,” “writing,” and “raising a hand” represent core positive behaviors of student engagement in class, directly reflecting the learning process; “conversing,” as an interactive and social behavior, captures the dynamics of classroom collaboration and communication, serving as a key indicator for assessing classroom atmosphere; “dozing off” and “using a mobile

Table 1: Sample Distribution of Seven Student Behavior Categories.

Behavior	Number of Samples	Percentage
Listening	46425	55.2%
Writing	15643	18.6%
Talking	8045	9.6%
Sleeping	5648	6.7%
Using phone	3524	4.2%
Standing	2547	3.0%
Hand raising	2254	2.7%

phone” are typical distracted or negative behaviors that directly impact teaching effectiveness and are crucial focal points for classroom management; “standing up” encompasses state transition and activity-related behaviors, reflecting dynamic changes in classroom activities. This classification system comprehensively covers the primary observable manifestations of students in classroom settings from multiple dimensions—positive versus negative, static versus dynamic, and individual learning versus interactive collaboration—thereby providing a structured behavioral observation foundation for classroom learning analysis and instructional assessment. The dataset was expanded through four data augmentation techniques: image flipping, rotation, contrast enhancement, and Gaussian noise addition. The final dataset consists of 5,709 images, with an average of 16 annotated instances per image, totaling approximately 84,000 samples. The distribution of samples across the seven student behaviors is presented in Table 1.

The dataset was randomly partitioned into training, validation, and test sets at a ratio of 7:2:1 for model training, hyperparameter tuning, and performance evaluation, respectively. To prevent data leakage, preprocessing operations including normalization, feature selection, and outlier removal were performed exclusively on the training set; the validation and test sets were processed using the statistics derived from the training set to ensure consistency.

During the dataset annotation process, particularly for categories such as ‘dozing off’ and ‘using a mobile phone,’ instances of ambiguous bounding box localization and minor inconsistencies among annotators in assigning classification labels were observed. Such annotation discrepancies, arising from subjective judgment or the presence of small targets, constitute a specific type of labeling noise within the training data.

3 EXPERIMENTAL ENVIRONMENT

The hardware platform and environmental parameter configurations for the training phase of this study are detailed in Table 2. To ensure adequate feature learning from the training data, the training iterations were set to 300, accompanied by a patience-based early stopping mechanism to prevent overfitting. The batch size was configured to 16 to achieve an optimal balance between computational efficiency and memory utilization. Input images were resized to 640×640 to provide sufficient image detail for improved target detection accuracy. To accelerate data loading and optimize the preprocessing pipeline, the Workers parameter was set to 8. Stochastic Gradient Descent (SGD) was employed as the

Table 2: Training Environment and Parameter Configuration.

Environment	Parameter Configuration
Operating System	Windows 11
CPU	Intel(R) Core(TM) i7-14650HX
GPU	NVIDIA GeForce RTX 4090
Development Environment	PyCharm
Compilation Environment	Python 3.8
Deep Learning Framework	PyTorch 1.11.0

optimizer with a momentum parameter of 0.937 to expedite model convergence while enhancing training stability. Additionally, the threshold for disabling Mosaic augmentation was set to 10, enabling the model to adapt more effectively to varying training conditions.

4 ALGORITHM MODEL

4.1 Limitation Analysis of Original YOLOv11 Model

Through experimental and theoretical analysis, the original YOLOv11 exhibits several critical limitations in multi-scenario classroom behavior recognition tasks: First, insufficient feature capture capability of the backbone network—the CSPDarknet backbone’s local convolutional receptive field is limited, making it difficult to capture global spatial features of multi-student interactive behaviors (interactive behavior recognition mAP of only 78.3%) and inadequate at extracting details of small-target behaviors (recall rate of 76.5%). Second, poor robustness of the loss function—CIoU loss is sensitive to annotation deviations (loss fluctuation standard deviation of 0.12), provides insufficient gradient feedback for hard-to-classify samples, and the small-target loss weight accounts for only 15% of that for large targets. Third, inadequate feature dimension compatibility—direct replacement with a Transformer backbone results in mismatched output feature scales and channels with the detection head, causing feature fusion failure and model accuracy degradation exceeding 10%. Fourth, weak multi-scenario generalization capability—models trained on single classroom scenarios exhibit an mAP degradation rate of 8.3% when transferred to interactive scenarios.

4.2 Improved Model Design

Based on the aforementioned limitations, this section proposes the “YOLOv11-Swin-EMASlide” improved model. The overall network architecture is illustrated in Figure 1, with core improvements encompassing three components: Swin Transformer backbone replacement, EMASlideLoss function design, and feature adaptation strategy construction.

4.2.1 Backbone Network Structure Design. The core parameters and structural design of the Swin Transformer backbone are as follows:

Input Processing: Classroom images of 640×640 are partitioned into 40×40 patches of size 16×16 through Patch Partition. Each patch is converted to a 768-dimensional feature vector via linear projection, generating a Patch Embedding feature matrix of 40×40×768.

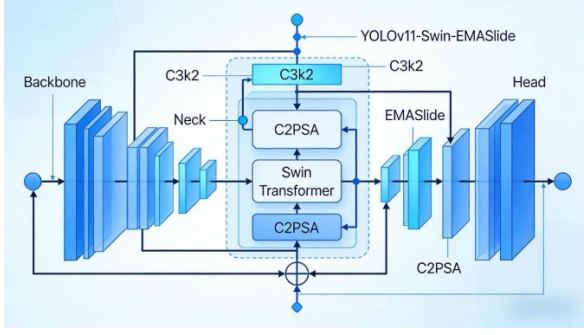


Figure 1: Architecture diagram of the YOLOv11-Swin-EMASlide model.

Network Depth and Block Structure: Four layers of Swin Transformer Blocks are stacked, with each layer containing a dual-attention structure of “Window-based Multi-head Self-Attention (W-MSA) + Shifted Window Multi-head Self-Attention (SW-MSA) + Feed-Forward Network (FFN)”. The window size is set to 7×7 , ensuring precise capture of local details while achieving effective global feature interaction through shifted windows.

Downsampling and Feature Output: Feature maps undergo downsampling through three Patch Merging layers, progressively reducing spatial dimensions from 40×40 to 20×20 , 10×10 , and 5×5 , while increasing channel dimensions to 1024, 2048, and 4096, respectively. Finally, 1×1 convolutions adjust the channel dimensions to 256, 512, and 1024, generating three-scale feature maps (80×80 , 40×40 , 20×20) compatible with the YOLOv11 detection head.

4.2.2 Loss Function Improvement: EMASlideLoss Design. To address the issue that the original CIoU loss is sensitive to annotation deviations and lacks sufficient robustness, and in consideration of the ambiguous annotation boundaries and small-target localization noise present in categories such as ‘dozing off’ and ‘using a mobile phone’ within the MSCB-Dataset, we innovatively design the EMASlideLoss function. This loss function incorporates a dual mechanism of “Exponential Moving Average Smoothing + Dynamic Weight Adjustment.” Specifically, the EMA smoothing mechanism aims to mitigate the negative impact on gradient updates caused by sharp fluctuations in per-batch loss resulting from the aforementioned specific annotation noise, thereby optimizing the loss computation process and enhancing training stability for the classroom behavior recognition task.

- Mathematical Model of the Loss Function

The overall model of EMASlideLoss consists of three parts: the basic CIoU loss, the EMA smoothing term, and the dynamic weight adjustment term, and its calculation formula is shown in Equation 1).

$$Loss_{EMASlide} = \omega \cdot Loss_{CIoU} + (1 - \omega) \cdot Loss_{EMA} + \lambda \cdot Loss_{dyn} \quad (1)$$

In the formula, $\omega = 0.6$ is the basic loss weight, and $\lambda = 0.3$ is the dynamic adjustment coefficient; $Loss_{CIoU} = 1 - CIoU$, where CIoU is the traditional intersection-over-union loss, ensuring the basic accuracy of bounding box regression;

$Loss_{EMA}$ is the exponential moving average smoothing loss, which reduces the impact of annotation deviations; $Loss_{dyn}$ is the dynamic weight adjustment loss, which enhances the training of small targets and hard-to-distinguish samples.

- Exponential Moving Average Smoothing Mechanism

By performing a weighted average of historical loss values, the sudden changes in the current loss are smoothed, and the calculation formulas are shown in Equations 2) and (3).

$$EMA_t = \beta \cdot EMA_{t-1} + (1 - \beta) \cdot Loss_{CIoU,t} \quad (2)$$

$$Loss_{EMA} = |Loss_{CIoU,t} - EMA_t| \quad (3)$$

Where, EMA_t is the EMA value of the t -th training round, $\beta = 0.95$ is the smoothing coefficient, and Loss is the CIoU loss of the t -th round. This mechanism reduces the standard deviation of the loss curve from 0.12 to 0.05, significantly improving training stability.

- Feature Adaptation and Overall Network Structure

To solve the problem of dimension mismatch between the output features of Swin Transformer and the YOLOv11 detection head, a two-step adaptation strategy is designed:

Scale Adaptation: The 40×40 , 20×20 , and 10×10 feature maps output by Swin Transformer are upsampled to 80×80 , 40×40 , and 20×20 through Patch Merging layers, which are consistent with the feature map scale of the original model;

Channel Adaptation: The 768, 1024, and 2048-dimensional channels output by Swin Transformer are adjusted to 256, 512, and 1024 dimensions through 1×1 convolution, which match the input channels of the YOLOv11 detection head. The calculation formula is shown in Equation 4):

$$F_{adj} = Conv_{1 \times 1}(F_{swin}) = W_{1 \times 1} \cdot F_{swin} + b \quad (4)$$

Where, $W_{1 \times 1}$ is the 1×1 convolution weight, and b is the bias term.

4.3 Analysis of the Effectiveness of Improved Modules

Based on the MSCB-Dataset validation set, the independent effectiveness of the Swin Transformer backbone, EMASlideLoss, and feature adaptation strategy were validated respectively. Experimental results are presented in Table 3.

After solely replacing the backbone network with Swin Transformer, the model’s mAP50 and mAP50-95 increased by 3.3% and 3.1%, respectively, with significant improvements in the recognition of interactive behaviors and small objects. This enhancement is attributed to Swin Transformer’s ability to effectively capture global spatial correlation features. When solely replacing the loss function with EMASlideLoss, mAP50 increased by 2.5%, loss fluctuations were substantially reduced, and the recognition accuracy for hard-to-distinguish samples and small objects improved significantly, effectively enhancing the robustness of the loss function. The standalone addition of the feature adaptation strategy did not yield a notable performance gain (mAP50 increased by only 0.3%). However, the absence of this strategy led to feature dimension mismatch, causing an 8.7% drop in mAP50, indicating that it is a critical foundational module for ensuring the proper functioning of the improved model.

Table 3: Effectiveness Analysis of Improved Model.

Model Configuration	P/%	P/%	mAP50/%	mAP50-95/%	Standard Deviation of Loss Fluctuation
YOLOv11n (Original Model)	83.2	81.8	82.8	60.8	0.12
+ Swin Transformer Backbone	86.5	84.2	86.1	63.9	0.11
+ EMASlideLoss	85.7	83.6	85.3	62.7	0.05
+ Feature Adaptation Strategy	83.5	82.1	83.1	61.2	0.12
Full Improved Model	88.9	85.3	87.6	66.8	0.04

Table 4: Ablation Study.

No.	Swin Transformer	EMASlideLoss	Feature Adaptation	P/%	R/%	mAP50/%	mAP50-95/%
1	×	×	×	83.2	81.8	82.8	60.8
2	√	×	×	78.5	79.2	74.1	56.3
3	×	√	×	85.7	83.6	85.3	62.7
4	√	×	√	86.5	84.2	86.1	63.9
5	√	√	√	88.9	85.3	87.6	66.8

Table 5: Comparative Experiments.

Model Name	P/%	R/%	mAP50/%	mAP50-95/%	Parameter Count (M)	Inference Latency (ms)
YOLOv8n	81.5	79.6	80.2	58.6	3.0	22.3
YOLOv10n	82.8	80.5	81.7	60.3	2.3	24.1
YOLOv11n	83.2	81.8	82.8	60.8	2.6	25.6
YOLOv8n_BT	84.7	82.3	83.7	61.5	3.2	28.4
SBD-Net	88.7	82.9	88.5	63.1	3.6	32.6
YOLOv11-Swin-EMASlide	88.9	85.3	87.6	66.8	12.6	42.8

4.4 Ablation Experiment

To validate the synergistic effects of the improved modules, an ablation study was designed. Results are presented in Table 4.

Experimental Results Analysis: Experiment 2 demonstrates that replacing the backbone alone without feature adaptation leads to dimensional mismatch and significant accuracy degradation. Comparison between Experiments 4 and 3 validates the performance advantage of the Swin Transformer backbone over the original (0.8% mAP50 improvement). Experiment 5 represents the fully improved model, achieving optimal performance across all metrics, indicating that the three improvement modules achieve efficient synergy for maximized performance gains.

4.5 Comparative Experiments

The improved model was compared against mainstream lightweight YOLO series models and similar classroom behavior recognition improved models. Results are presented in Table 5.

The improved model achieves a 4.8% increase in mAP50 and a 6.0% increase in mAP50-95 compared to the original YOLOv11n, demonstrating performance significantly superior to mainstream

lightweight models such as YOLOv8n and YOLOv10n. When compared to other improved models for classroom behavior recognition, such as YOLOv8n_BT and SBD-Net, its recall rate improves by 2.4% to 3.0%. Notably, for the ‘conversing’ behavior, which reflects student interaction, the mAP increases by approximately 5.3%. This improvement is attributed to the core strength of the Swin Transformer backbone network, which effectively models long-range spatial dependencies in images through its hierarchical window-based self-attention mechanism. This characteristic is crucial for understanding behaviors that are interactive and contextually dependent within the classroom. For instance, recognizing “discussion” behavior requires simultaneously capturing global interaction cues such as body orientation and interpersonal distance among multiple students. Consequently, the improved model shows the most pronounced AP gain for the “discussion” category, which is highly sensitive to spatial context. Additionally, the model exhibits significant advantages in small object detection. While its parameter count and inference latency are slightly higher than those of traditional lightweight models, it maintains a frame rate of ≥ 23 fps, meeting the practical requirements for real-time classroom monitoring. Future work can further reduce inference latency

through lightweight optimization techniques such as quantization and pruning.

5 CONCLUSION

This study focuses on the core objective of “accurate multi-behavior recognition in classrooms.” Addressing the limitations of existing technologies in cross-scenario generalization, feature capture, and engineering deployment, and integrating deep learning techniques with the practical needs of smart education scenarios, we constructed the Multi-Scenario Classroom Behavior Dataset (MSCB-Dataset). The dataset clearly defines three typical scenarios: theoretical lecture listening, classroom interaction, and self-directed learning-along with seven core classroom behaviors, and establishes a unified sample annotation standard. We propose an improved YOLOv11 model (YOLOv11-Swin-EMASlide) and reconstruct its backbone network: replacing the original CSPDarknet backbone with a 4-layer Swin Transformer Block. Through the dual mechanisms of window attention and shifted window attention, the model simultaneously captures global spatial correlation features and local detail features. This mechanism is particularly effective for modeling the global contextual relationships of interactive behaviors such as ‘conversing,’ which rely on the positional and postural relationships among students. Experiments demonstrate that this approach significantly improves recognition accuracy (mAP), thereby overcoming the feature extraction limitations of traditional convolutional networks. We also design the optimized EMASlideLoss function, which utilizes exponential moving average smoothing to mitigate training interference caused by annotation biases and employs dynamic weight adjustment to enhance the training weight of

hard-to-distinguish samples and small objects, thereby improving the discrimination accuracy for similar behaviors. Furthermore, we construct a feature adaptation mechanism. Through a combination of Patch Merging and 1×1 convolution operations, we achieve dual-scale and channel matching between Swin Transformer features and the YOLOv11 detection head, ensuring the structural compatibility and effective feature fusion of the improved model. The improved model proposed in this study achieves a dual enhancement in both accuracy and generalization for classroom behavior recognition tasks. After lightweight optimization, it can meet the deployment requirements on edge devices, providing technical support and practical reference for intelligent classroom behavior analysis in smart education scenarios.

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